UNIVERSITY of HOUSTON

DEPARTMENT OF COMPUTER SCIENCE

Security Analytics

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Lecture 6: Intrusion Detection Systems

(Ref: Slides adapted from those accompanying book by William Stallings)

Definitions

Intrusion

A set of actions aimed to compromise the security goals, namely

Integrity, confidentiality, or availability, of a computing and networking resource

Note: This is a very broad definition, usually we refer to a penetration of the system as an intrusion (informal)

Intrusion detection

The process of identifying and responding to intrusion activities

Elements of Intrusion Detection

Primary assumptions:

System activities are observable

Normal and intrusive activities have distinct evidence

Components of intrusion detection systems:

From an algorithmic perspective:

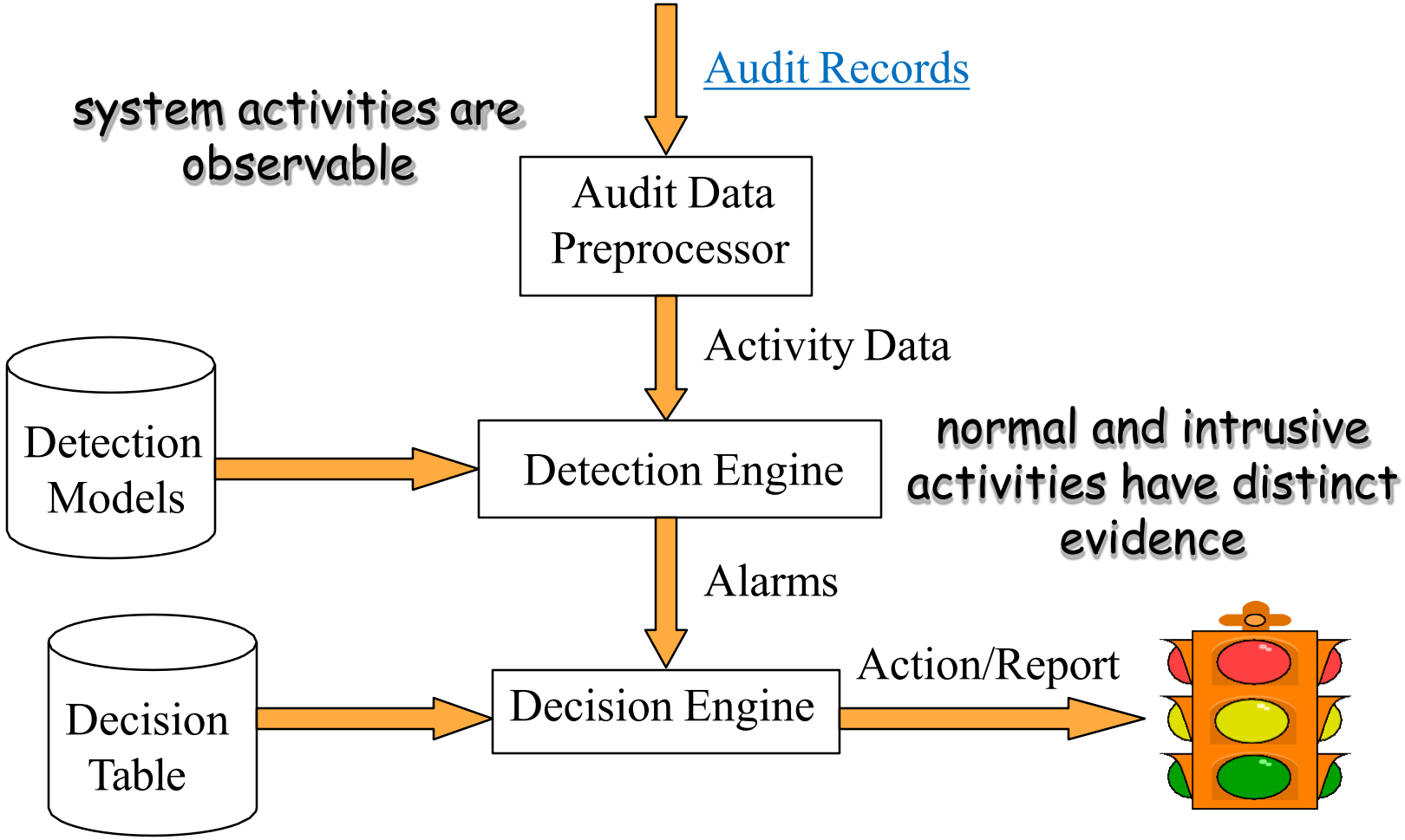
Features - capture intrusion evidences

Models - piece evidences together

From a system architecture perspective:

Audit data processor, knowledge base, decision engine, alarm generation and responses

Components of Intrusion Detection System



Intrusion Detection Approaches

Modeling

Features: evidences extracted from audit data

Analysis approach: piecing the evidences together

Misuse detection (a.k.a. signature-based)

Anomaly detection (a.k.a. statistical-based)

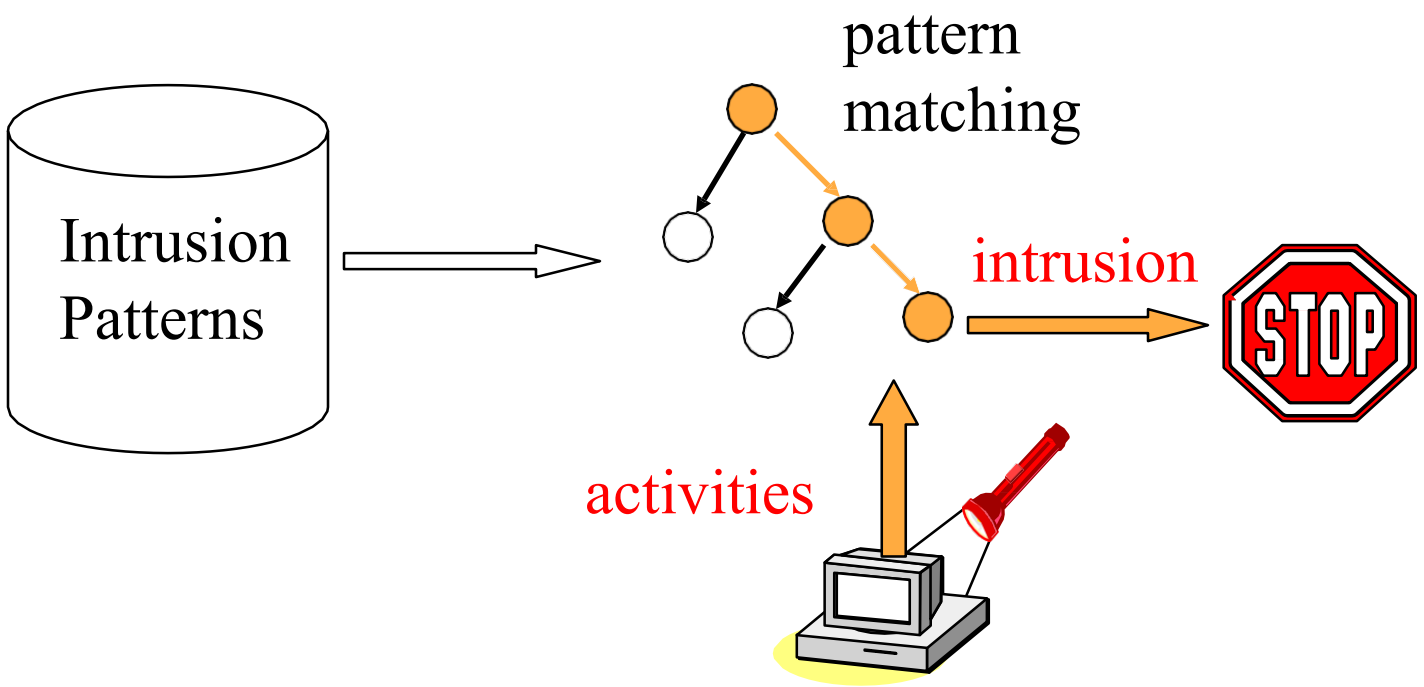
Deployment: Network-based or Host-based

Development and maintenance

Hand-coding of “expert knowledge”

Learning based on audit data

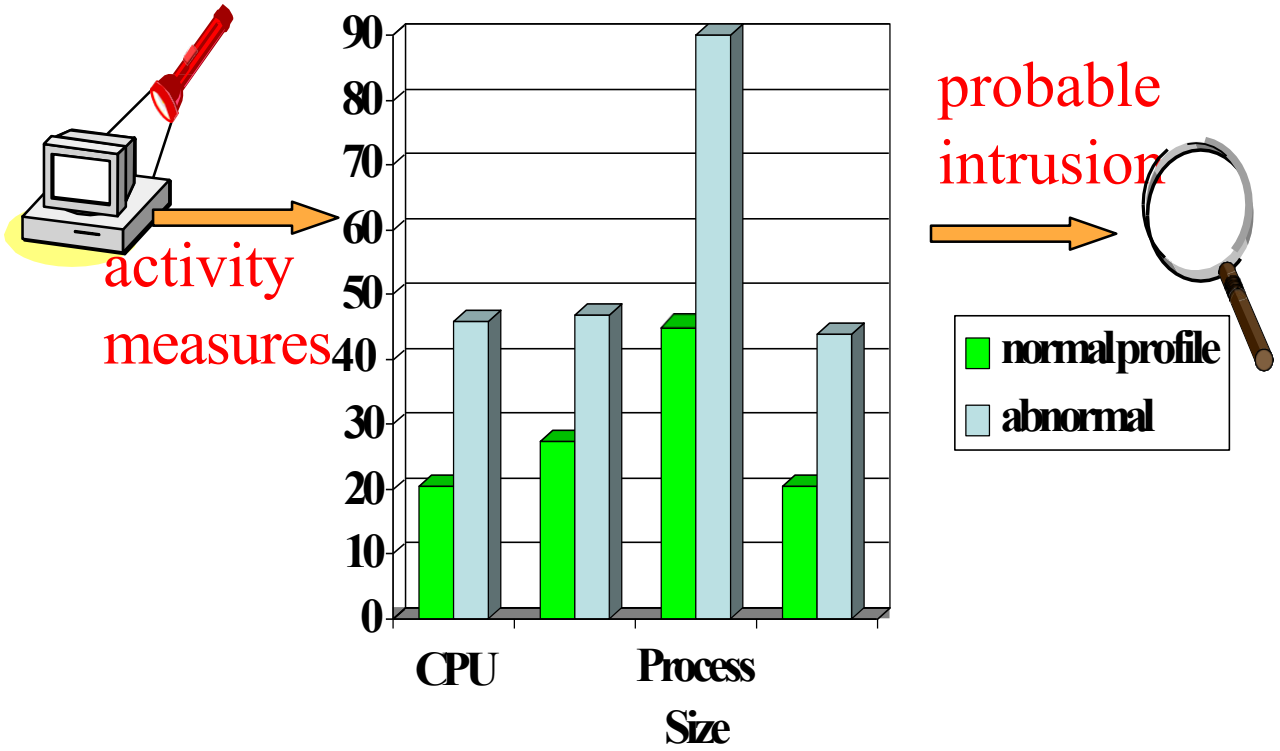
Misuse Detection



Example: ***if*** (src_ip == dst_ip) ***then*** "land attack"

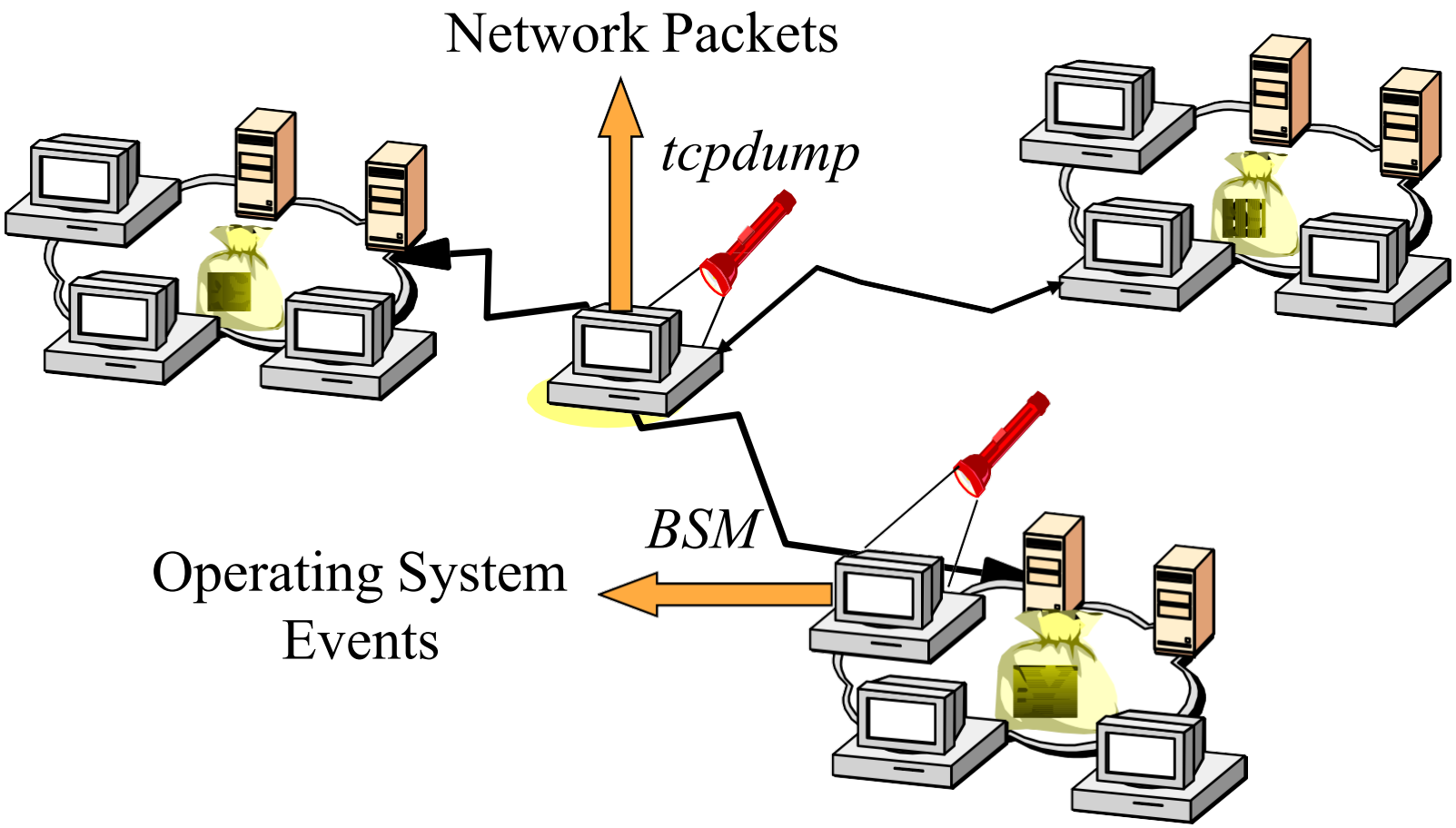
Can't detect new attacks

Anomaly Detection



Relatively high false positive rate -
anomalies can just be new normal activities.

Monitoring Networks and Hosts



Key Performance Metrics

Algorithm

Alarm: A; Intrusion: I, we call Intrusion the positive class and normal activity the negative class

Detection (true alarm) rate: $P(A|I)$, True positives

False negative rate $P(\neg A|I)$

False alarm rate: $P(A|\neg I)$

True negative rate $P(\neg A|\neg I)$

Architecture

Scalable

Resilient to attacks

Host-Based IDSs

Using OS auditing mechanisms

E.G., BSM on Solaris: logs all direct or indirect events generated by a user

strace for system calls made by a program

Monitoring user activities

E.G., Analyze shell commands

Monitoring executions of system programs

E.G., Analyze system calls made by *sendmail*

Network IDSs

Deploying sensors at strategic locations

E.G., Packet sniffing via *tcpdump* at routers

Inspecting network traffic

Watch for violations of protocols and unusual connection patterns

Monitoring user activities

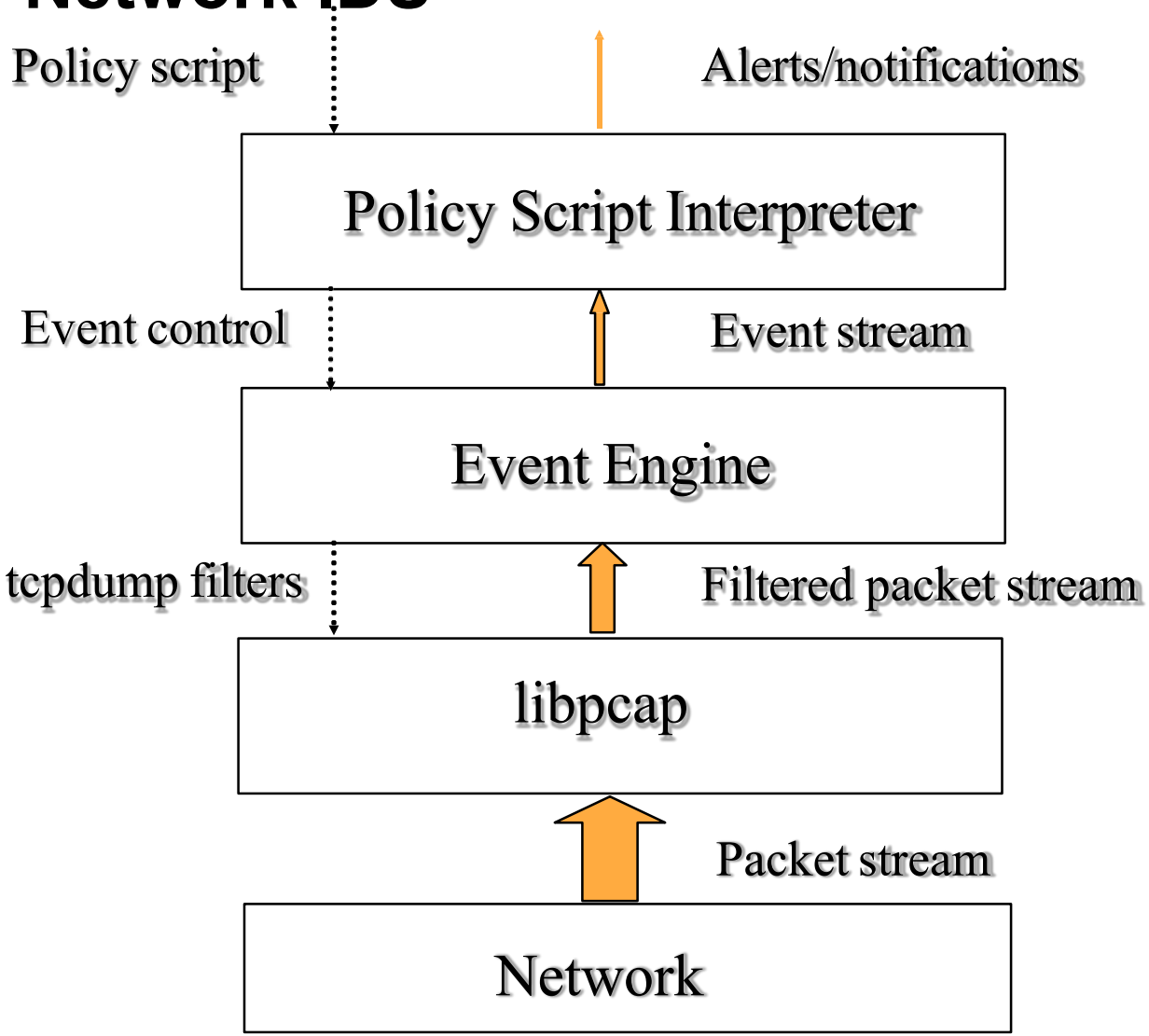
Look into the data portions of the packets for malicious command sequences

May be easily defeated by encryption

Data portions and some header information can be encrypted

Other problems …

Architecture of Network IDS



Firewall Versus Network IDS

Firewall

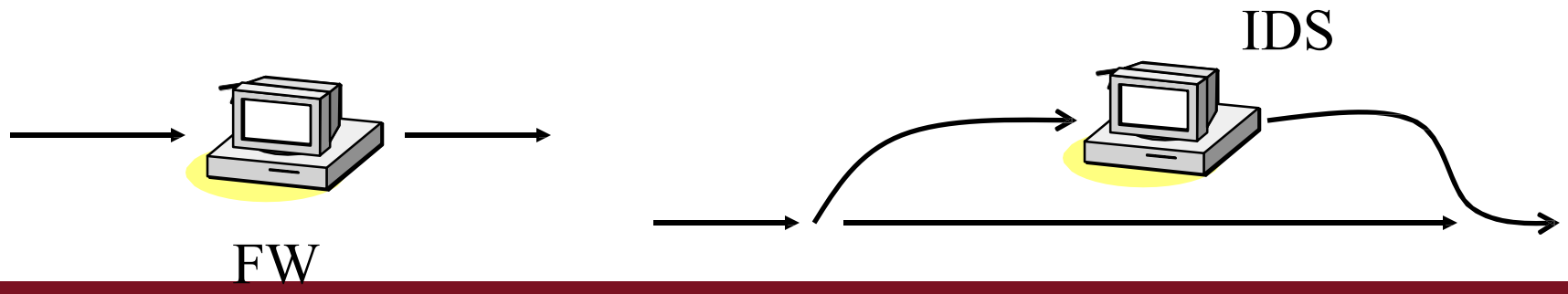
Active filtering

Fail-close

Network IDS

Passive monitoring

Fail-open



Requirements of Network IDS

High-speed, large volume monitoring

No packet filter drops

Real-time notification

Mechanism separate from policy

Extensible

Broad detection coverage

Economy in resource usage

Resilience to stress

Resilience to attacks upon the IDS itself!

Case Study: Snort IDS

Problems with Current IDSs

Knowledge and signature-based:

"We have the largest knowledge/signature base"

Ineffective against new attacks

Individual attack-based:

"Intrusion *A* detected; Intrusion *B* detected …"

No long-term proactive detection/prediction

Statistical accuracy-based:

"*x*% detection rate and *y*% false alarm rate"

Are the *most damaging* intrusions detected?

Statically configured.

Next Generation IDSs

Adaptive

Detect new intrusions

Scenario-based

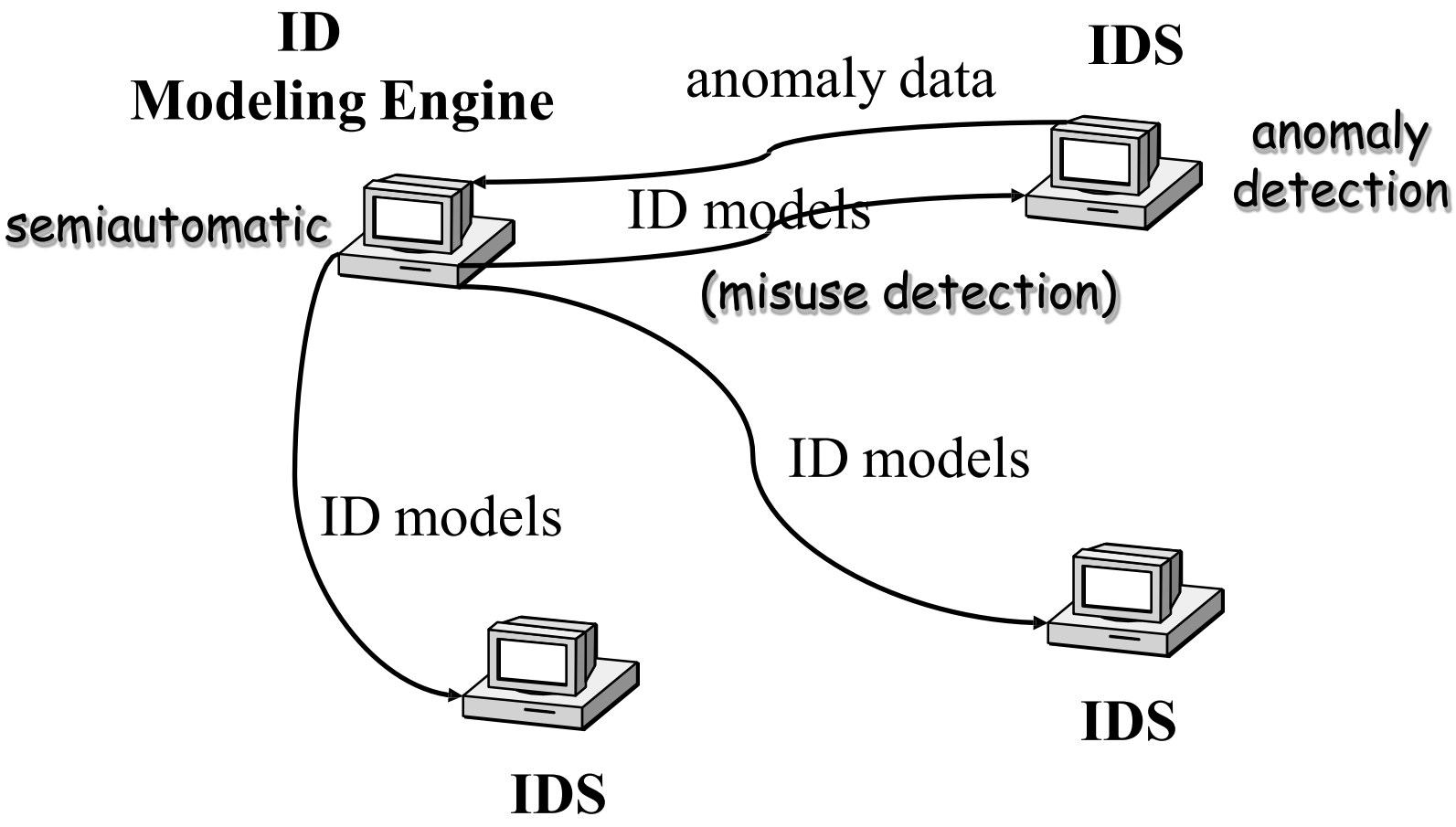
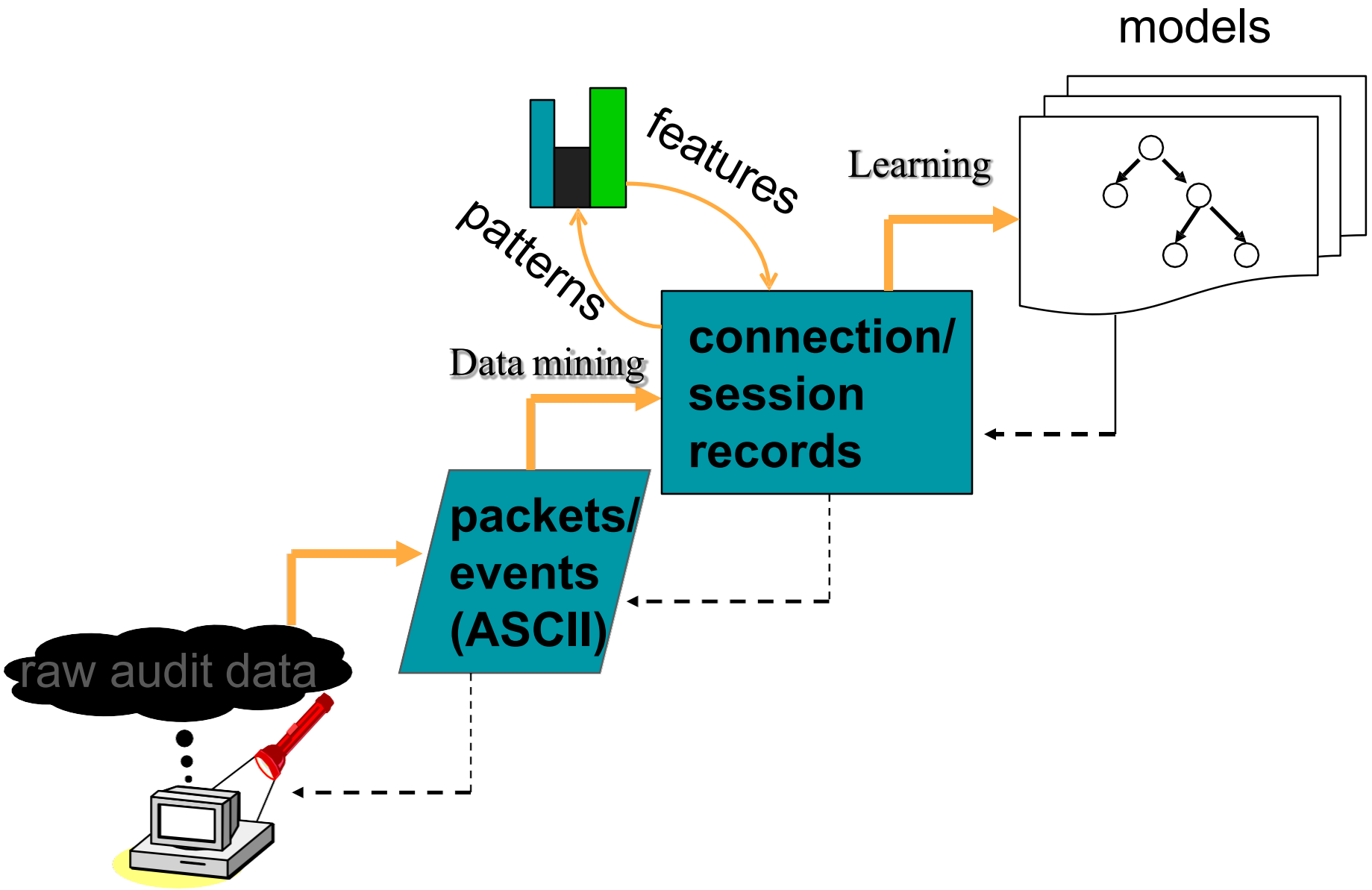
Correlate (multiple sources of) audit data and attack information

Cost-sensitive

Model cost factors related to intrusion detection

Dynamically configure IDS components for best protection/cost performance

Adaptive IDSs



Semi-automatic Generation of ID Models